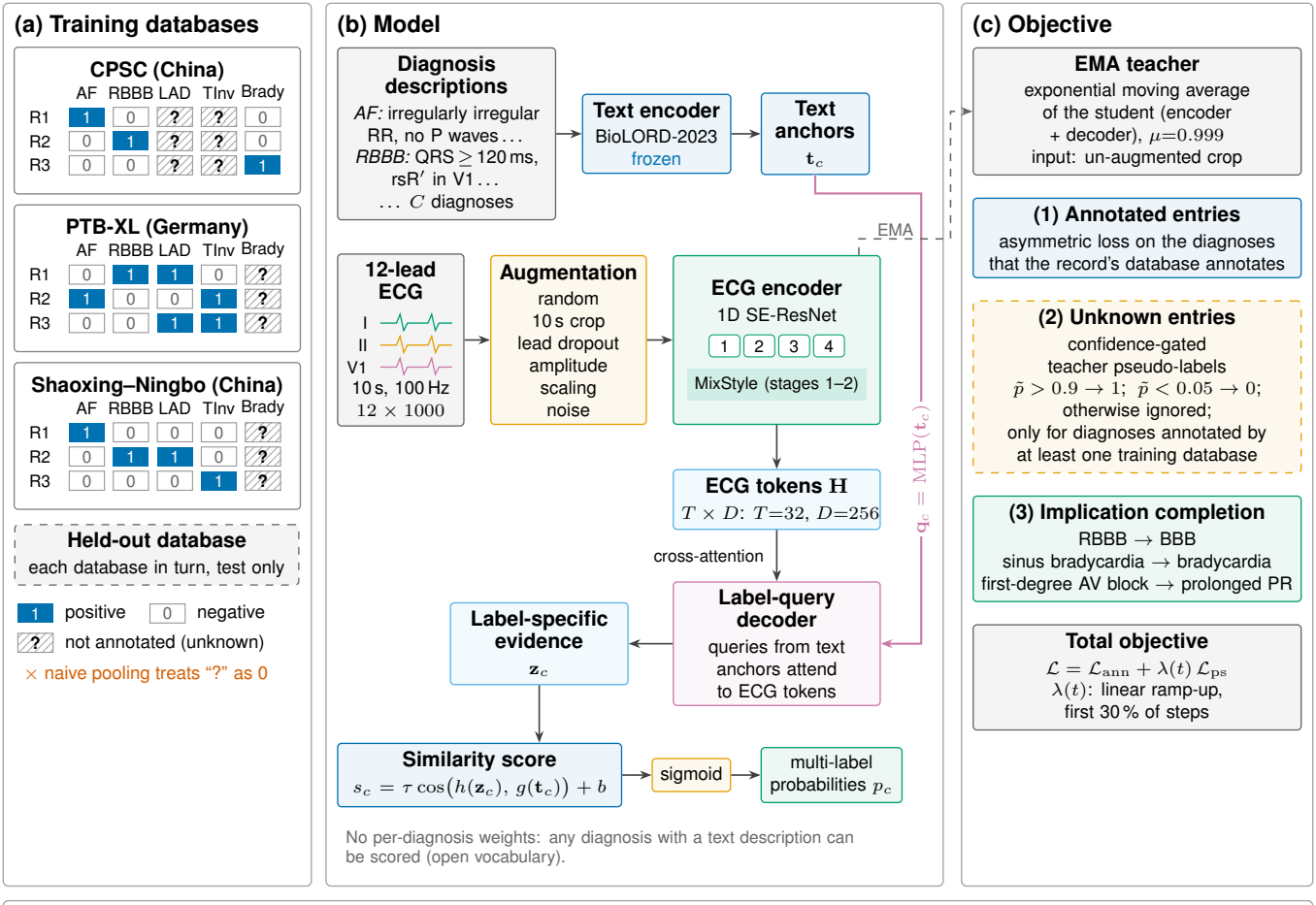

(a) Training databases
CPSC (China)
AF RBBB LAD TInv Brady
R1 1 0 ? ? 0
R2 0 1 ? ? 0
R3 0 0 ? ? 1
PTB-XL (Germany)
AF RBBB LAD TInv Brady
R1 0 1 1 0 ?
R2 1 0 0 1 ?
R3 0 0 1 1 ?
Shaoxing–Ningbo (China)
AF RBBB LAD TInv Brady
R1 1 0 0 0 ?
R2 0 1 1 0 ?
R3 0 0 0 1 ?
Held-out database
each database in turn, test only
1 positive
0 negative
? not annotated (unknown)
× naive pooling treats "?" as 0
(b) Model
Diagnosis descriptions
AF: irregularly irregular RR, no P waves . . .
RBBB: QRS ≥ 120 ms, rsR′ in V1 . . .
. . . C diagnoses
Text encoder
BioLORD-2023
frozen
Text anchors
$\mathbf{t}_c$
EMA
12-lead ECG
I
II
V1
10 s, 100 Hz
$12 \times 1000$
Augmentation
random 10 s crop
lead dropout
amplitude scaling
noise
ECG encoder
1D SE-ResNet
1 2 3 4
MixStyle (stages 1–2)
$\mathbf{q}_c = \mathrm{MLP}(\mathbf{t}_c)$
ECG tokens $\mathbf{H}$
$T \times D$: $T$=32, $D$=256
cross-attention
Label-query decoder
queries from text anchors attend to ECG tokens
Label-specific evidence
$\mathbf{z}_c$
Similarity score
$s_c = \tau \cos\big(h(\mathbf{z}_c),\, g(\mathbf{t}_c)\big) + b$
sigmoid
multi-label probabilities $p_c$
No per-diagnosis weights: any diagnosis with a text description can be scored (open vocabulary).
(c) Objective
EMA teacher
exponential moving average of the student (encoder + decoder), $\mu$=0.999
input: un-augmented crop
(1) Annotated entries
asymmetric loss on the diagnoses that the record's database annotates
(2) Unknown entries
confidence-gated teacher pseudo-labels
$\tilde{p} > 0.9 \rightarrow 1$; $\tilde{p} < 0.05 \rightarrow 0$;
otherwise ignored;
only for diagnoses annotated by at least one training database
(3) Implication completion
RBBB → BBB
sinus bradycardia → bradycardia
first-degree AV block → prolonged PR
Total objective
$\mathcal{L} = \mathcal{L}_{\mathrm{ann}} + \lambda(t)\, \mathcal{L}_{\mathrm{ps}}$
$\lambda(t)$: linear ramp-up, first 30 % of steps
Evaluation
E1 leave-one-database-out (seen and unseen diagnoses)
E2 controlled annotation removal (PTB-XL)
Ablations learned queries vs. text anchors; loss and augmentation